The Assembly: Null Lines Through a Point, the Light-Cone Identification, and Lorentz Symmetry as Output

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Abstract. The geometrization paper ended with a proposal: identify the thermofield shadow of each radial ray with its antipodal ray, and assemble the per-direction chiral systems over the sphere. This note executes the assembly, resolving its conceptual hurdles first and letting the computations correct the proposal where it needed correcting. The architectural decision — extend by *identification*, not by tensoring — is made on canonicity: the antipodal ray algebra is represented on the existing GNS space as the commutant, adding no Hilbert space and no state data, so that each axis carries the axis *vacuum*, pure and Unruh-entangled across its rays, restricting to exactly the series' thermal state on each. Mutual locality of the family is exact (disjoint angular supports give vanishing symplectic form), so the axis systems commute and assemble, with rotations permuting them. The recognition that decides the phase: a family of chiral lines over the direction sphere, each carrying $PSL(2,\mathbb{R})$, is the kinematic arrangement of a *light cone* — and the identification is then verified against 3+1 Minkowski space directly. Restricting the massless vacuum's Wightman function to the cone $x = \lambda(1, \hat{n})$ and forming the current data $\partial_\lambda(\lambda\phi)$: cross-generator correlations vanish *identically* (verified at 10^{-19}; the undifferentiated correlation is constant along generators off-diagonal), and the sphere-integrated same-generator kernel is exactly $-1/4\pi(\lambda - \mu - i\epsilon)^2$ — the series' chiral vacuum kernel as a boundary value, constant for constant, verified to 4×10^{-6}. The assembled state *is* the light-cone current data of the Minkowski massless vacuum. The computation also corrects the pairing: the cross-generator zero holds at $c = -1$ too, so the shadow of a future-cone generator is not the same cone's antipodal generator but the corresponding *past*-cone generator — the axis is the complete null geodesic through the apex, and the modular conjugation maps future to past, its PT flavor now geometric. Consequence: Lorentz transformations preserve the null-line family and the Minkowski vacuum, so the assembled state is *exactly* Lorentz invariant — the symmetry arriving as output, with no mechanism for approximate violation, precisely as the series' Lorentz audit required. The soft sector — the cross-generator, zero-mode correlations the current condition excludes — is named as the standing frontier.

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1. The hurdles, resolved in order

A. Identification, not tensoring. The series' state lives on ray algebras; the antipodal proposal concerns axes. Two extensions exist: adjoin the antipodal ray as an independent tensor factor (two uncorrelated heat baths per axis), or identify it with the thermofield shadow already present. The geometrization paper decides this: the GNS space of the ray-thermal state *is* the axis vacuum space, with the shadow $B(r < 0)$ already represented as the commutant. Identification is the canonical, minimal extension — no new Hilbert space, no new state data — and under it each axis carries the axis vacuum: pure, entangled across its rays in the Unruh pattern, restricting to exactly the series' thermal state on each ray. The shadow is not a second world; it is the other half of this one.

B. Mutual locality. Smearings with disjoint angular supports have vanishing symplectic form — the direction overlap factorizes out (verified: overlap 0.0 for disjoint caps) — so axis systems along different directions commute, and the family assembles with $SO(3)$ permuting the axes.

C. The recognition. A family of chiral lines over the sphere of directions, each carrying a positive-energy $PSL(2,\mathbb{R})$, with rotations acting on the family: this is the kinematic arrangement of a *light cone,* generators labeled by the celestial sphere. The higher-dimensional question thereby becomes checkable: is the assembled state the light-cone data of the 3+1 Minkowski massless vacuum?

2. The light-cone identification, verified

Restrict the massless Wightman function to the cone, $x = \lambda(1,\hat{n}), y = \mu(1,\hat{m}), c = \hat{n}\cdot\hat{m}$:

$$(x-y)^2 = -2\lambda\mu(1-c), \qquad W_\psi := \lambda\mu\, W = \frac{1}{4\pi^2}\,\frac{\lambda\mu}{2\lambda\mu(1-c)+2i\epsilon(\lambda-\mu)+\epsilon^2},$$

with $\psi = \lambda\phi$ the null-projected (conformally rescaled) field and $\partial_\lambda\psi$ the current data.

Proposition 1 (generator-diagonality). For distinct generators ($c<1$), $W_\psi = 1/8\pi^2(1-c)$ is *constant* in λ, μ as $\epsilon \to 0$, so the cross-generator current correlation $\partial_\lambda\partial_\mu W_\psi$ vanishes identically — a one-line fact. Verified numerically at $c = 0.3,\ -0.45,\ 0.8$: residuals $5\times10^{-20}, 9\times10^{-13}, 4\times10^{-19}$.

Proposition 2 (the kernel). The sphere-integrated same-generator current kernel is exactly the series' chiral vacuum kernel. Analytically: $\int d\Omega\, W_\psi = (1/4\pi)\ln[(2+i\eta)/i\eta]$ with $\eta = \epsilon(\lambda-\mu)/\lambda\mu$, whose mixed λ,μ-derivative as $\epsilon\to0$ is the boundary value $-1/4\pi(\lambda-\mu-i\epsilon)^2$ — the constant included, and the $i\epsilon$ retained: the prescription, inherited from the Wightman function through η, is what carries the antisymmetric part of the two-point function, without which the commutator — and with it the CCR content — would vanish. Off the diagonal the identity holds as functions, which is what the numerics test. Verified numerically (endpoint layer resolved on a logarithmic grid, differentiation at integrand level): relative deviations $2.2\times10^{-5}, 4.0\times10^{-6}, 2.2\times10^{-5}$ at three (λ,μ) pairs.

Conclusion. The assembled state of §1 — generator-diagonal, per-generator chiral vacuum with kernel $-1/4\pi(\lambda-\mu-i\epsilon)^2$ — coincides, structure for structure and constant for constant, with the light-cone current data of the 3+1 Minkowski massless vacuum.

3. The pairing, corrected by the computation

The geometrization paper's proposal paired a ray with the antipodal generator of the *same* cone. Proposition 1 refutes that reading: the cross-generator zero holds at $c = -1$ as well — antipodal future generators are uncorrelated in the current sector, so they cannot be each other's thermofield purification. The correct pairing is fixed by the dilations themselves: the scaling flow $x \mapsto e^t x$ preserves the *complete null geodesic* $s \mapsto s(1, \hat{n})$, $s \in \mathbb{R}$, whose two rays are the future-cone generator along $\hat{n}$ ($s > 0$) and the *past*-cone generator, antipodally placed ($s < 0$); restricted to such a line, the Minkowski vacuum is the full-line chiral vacuum, and the ray split is the Bisognano-Wichmann wedge split — thermal on each ray, entangled across them, exactly the axis structure of §1. **The axis is the complete null geodesic through the apex; the shadow of a future generator is its past continuation; and the modular conjugation maps future data to past data** — the PT flavor the reflection Θ displayed on the scale line, now realized geometrically as the exchange of the two cones. The antipodal identification stands, refined: antipodal in direction *and* in time orientation.

4. Lorentz symmetry as output

Lorentz transformations preserve the set of null lines through the apex, acting on each line by affine (Möbius-subgroup) rescaling and on the direction sphere by conformal transformations; and they preserve the Minkowski vacuum. By §2, the assembled state is that vacuum's cone data; therefore **the assembled state is exactly Lorentz invariant.** This is the fulfilment, at the level the assembly reaches, of the series' Lorentz audit: the architecture was shown to have no mechanism for small violations — symmetry would arrive exact or not at all — and it has now arrived, exact, as *output*: nothing Lorentzian was put in, and the invariance is inherited through an identification verified numerically against the constant. Per-line, the ingredients were all the series' own: the flow from the octonionic dilation's $\hbar$-preserving shadow, the state unique by thermality, the Möbius group generated from three modular flows, the isotropy of the family the G_2 remnant. What is claimed is the symmetry of the assembled *state and kinematics*; the reconstruction of bulk fields and dynamics in the cone's interior — holography in the established sense — is the next phase, not this one.

5. The soft sector: the standing frontier

The undifferentiated boundary correlation $W_\psi = 1/8\pi^2(1-c)$ — constant along generators, nontrivial *across* them — is precisely the sector the current condition removes: the zero modes. This is not a discard to forget: in the modern literature that sector is where memory and the infrared structure of massless theories live. The series' infrared condition, having served finiteness, necessity, uniqueness, and locality, here acquires its sharpest characterization: it is the restriction to the *hard* sector of the cone, generator-diagonal and Lorentz-covariant, with the soft, cross-generator sector excluded — named now as the standing frontier for the phase in which charges, memory, and eventually interactions are addressed.

6. Scope

Five items. *(i)* The extension of §1 is a definition, argued from canonicity; its vindication is §2's identification, which is verified at the two-point/current level — higher correlations are fixed by quasi-freeness on both sides, but that inference is stated, not re-verified. *(ii)* Lorentz invariance is established for the state and the null-line kinematics; unitary implementation on the assembled GNS space follows from the standard vacuum representation and is cited, not re-derived. *(iii)* Bulk reconstruction from cone data (the holography step, in the tradition of rigorous null-surface holography) is the next phase; nothing about the cone's interior is claimed here. *(iv)* The soft sector of §5 is excluded by construction, not resolved; charges and memory belong to it. *(v)* The time-orientation content of §3 — the conjugation exchanging the cones — is stated as geometry, and its thermodynamic reading (the state-selected arrow, in the series' earlier sense) is left as the interpretive adjacency it is.

References

- Companion papers: *Geometrization: The Exponential Correspondence, Thermal Relative Duality, and the Shadow's Home*; *Localization on the Scale Line: The Local Current Net, the Thermal Wedge, and the Modular Seed of the Möbius Group*; *The Squeeze-Thermal State: A Type-I No-Go, the Scale-Line Construction, and the Type III_1 Verdict*; *The State Is Unique: Quasi-Free Rigidity, the Closed Zero-Mode Door, and the Infrared Condition's Third Duty*; *The Second Contraction: From Algebraic R-Flux to the Heisenberg Algebra, the Dilation Symmetry It Creates, and the SO(3) Remnant of G2.*
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